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Business Report

Introduction

When a borrower fails to repay a loan (default), the bank may recover part of the outstanding amount through repayments, collections, or other recovery measures(recoveries). However, some of the amount may remain unrecovered, resulting in a financial loss for the bank(Loss, given default occurs). Understanding the likely extent of these losses is important for managing credit risk and planning for potential future losses.

The objective of this project is to develop a framework that estimates the proportion of a defaulted loan that may not be recovered. The analysis uses historical loan data to identify patterns in recovery outcomes and assess how reliably these patterns can be used to estimate future losses.

The framework is intended to support the bank's understanding of its loan portfolio. It can help risk managers compare different groups of loans, assess the possible effect of changes in portfolio composition, and identify areas where closer monitoring may be required.

Key Findings

The analysis provides several useful observations about the portfolio and the reliability of the estimates.

Overall loss estimates were close to the historical outcomes. Across the loans used for testing, the framework estimated an average loss rate of approximately 93.8%, compared with an observed average of approximately 93.9%. This indicates that the model captured the overall level of losses reasonably well in the test sample.

The accuracy of individual estimates remains a concern. Although the average estimates were close to the observed results, the model was less successful at capturing the differences between individual loans. Actual loss rates varied considerably, while the model's predictions were concentrated within a relatively narrow range. Therefore, an estimate that works well at the portfolio level may still differ substantially from the eventual outcome for a particular loan.

Performance differed across credit grades. The model produced different levels of prediction error across the seven credit grades. Errors were higher for some groups, particularly grade F, while other groups showed lower errors. This suggests that a single overall performance measure does not tell the complete story. The bank would need to examine results by segment before relying on the estimates across its entire portfolio.

Changes in portfolio composition affected estimated losses. The scenario analysis examined what would happen if the proportion of loans in different credit-grade groups changed. Increasing the combined share of grades A–C by ten percentage points increased the estimated portfolio loss rate by approximately 0.21 percentage points. Increasing the share of grades E–G produced a decrease of approximately 0.21 percentage points in the estimated loss rate.

These results reflect the recovery patterns observed in this particular dataset. They should not be interpreted as evidence that one group of credit grades is inherently riskier than another. The findings instead demonstrate why portfolio composition should be considered when assessing expected recoveries.

Implications for the Bank

The framework could provide a useful additional source of information for the bank's credit risk team. Portfolio-level estimates can help management understand the potential scale of losses following default, while comparisons across credit grades can highlight areas where recovery performance deserves closer attention.

The analysis also shows the importance of looking beyond the overall portfolio average. Two portfolios may have different estimated loss rates because they contain different proportions of loans with different historical recovery patterns. Monitoring these changes can help the bank understand how its portfolio is evolving and whether its expectations of future recoveries need to be reviewed.

However, the results also indicate that the framework should not be used in isolation. Its estimates are more informative when considered alongside actual recovery experience, information about individual borrowers, collection performance, and the bank's existing credit risk assessments.

Limitations and Areas for Improvement

The framework is based on historical observations. Recovery outcomes may change as economic conditions, borrower behaviour, collection practices, and the composition of the loan portfolio change. Consequently, patterns observed in the historical data may not continue in the future.

Another limitation is the difference between estimating the overall loss rate and predicting the outcome of an individual loan. The framework performed reasonably well in estimating the portfolio average, but its predictions did not fully reflect the wide variation in actual losses. This limits the extent to which the estimates can be used to make decisions about individual defaulted loans.

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The analysis also highlights the need to review performance across different segments. An acceptable overall result may conceal weaker performance in particular groups. Regular monitoring by credit grade and other relevant portfolio characteristics would help identify such weaknesses.

Before the framework is used in routine business decisions, the bank should test it on more recent data, compare its estimates with actual recovery outcomes, and assess whether its performance remains stable over time. The bank should also confirm that all information used by the model would be available at the time an estimate is required.

Recommendations

Based on the findings, the following actions are recommended:

- Use the framework as a supporting tool. Treat its estimates as additional information for risk assessment rather than as definitive predictions of future losses.
- Monitor actual versus estimated outcomes. Compare model estimates with realised recoveries regularly to identify changes in performance.
- Review individual portfolio segments. Investigate groups with larger prediction errors and determine whether improvements are needed.
- Assess changes in portfolio composition. Revisit loss estimates when the distribution of loans across credit grades or other relevant groups changes materially.
- Conduct further validation. Test the framework on more recent data and evaluate whether its performance remains reliable under different conditions.

Conclusion

This project demonstrates how historical loan information can help a bank estimate the proportion of defaulted loans that may remain unrecovered. The framework produced an overall loss estimate close to the observed portfolio average and performed slightly better than a simple historical benchmark. At the same time, the analysis identified limitations in predicting individual outcomes and differences in accuracy across credit grades.

The findings suggest that the framework has potential value as a portfolio monitoring and risk assessment tool. Its practical usefulness will depend on continued validation, regular monitoring, and careful interpretation of the results. With these safeguards, it can help the bank develop a clearer understanding of recovery performance and make better-informed decisions about managing losses following default.